

Price construction update

September 29, 2026 | Current specification

This page updates price construction in the attached document. Original pages follow unchanged. The definition below supersedes any earlier firm-price aggregation convention.

$$p_{\text{UPC},t} = \text{real revenue}_{\text{UPC},t} / \text{mL sold}_{\text{UPC},t}$$

$$p_{i,t} = \text{simple mean of } p_{\text{UPC},t} \text{ across eligible UPCs in firm } i, \text{ month } t$$

First collapse multiple T2 rows to one brand x UPC x month observation, summing real revenue and mL. Then calculate each UPC price and average across the firm's eligible UPCs, giving each UPC equal weight.

Use the existing sample: UNKNOWN excluded, positive units and mL, and at least three reporting stores per UPC-month. Price remains in real dollars per mL. The across-UPC weighting matches the UPC-unweighted nicotine state $a_{i,t}$.

Quantities, revenue totals, shares, market size and the population-weighted national tax instrument retain their existing construction. The descriptive fringe p_F remains aggregate fringe revenue / mL; it is not a price regressor or strategic price in this model.

The product-line average price times total firm mL generally does not equal observed revenue. Model revenue and profit using this representative price are not an accounting reconstruction of scanner revenue.

Implemented in code/build/8_build_firm_month_demand_panel.R. See docs/price_construction_update.md for the editable specification. This update addresses price construction only; it does not resolve other open modeling choices.